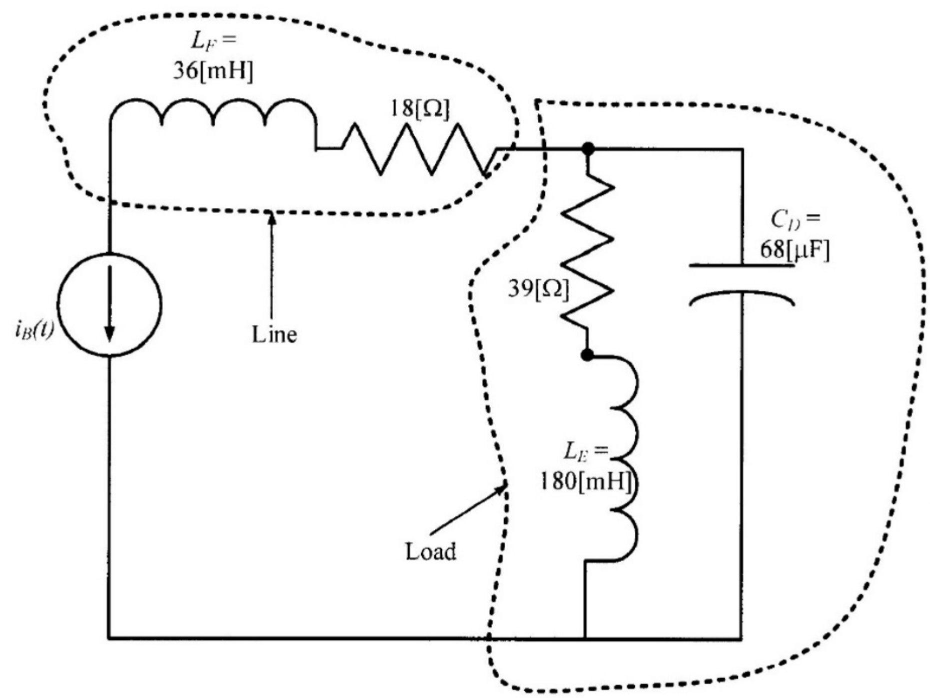
3. {30 Points} The circuit shown below operates in steady state.
a) Find the real power absorbed by the load.
b) Find the reactive power delivered by the load.
c) Find the complex power absorbed by the load.
d) Find the power factor for the load, including whether it is leading or lagging.
e) Find the susceptance of the line.

$$i_B(t) = 530[\mathrm{A}]\cos\left(420\left[\frac{\mathrm{rad}}{\mathrm{s}}\right]t - 38^\circ\right)$$



$$Z_{LOAD} = \frac{(39 + 75.6j)(-35.014j)}{39 + (75.6 - 35.014)j}\,\{\Omega\} = (52.917\angle -73.43^\circ)\{\Omega\}$$

$$Z_{LOAD} = (15.091 - 50.719j)\{\Omega\}$$

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$$\left|\bar{I}_{bm}\right| = 530\{A\}$$ We can use this value to find what we want.

a) $$P_{abs.by.load} = \frac{\left|\bar{I}_{bm}\right|^2 R_{LOAD}}{2} = \frac{(530\{A\})^2(15.091\{\Omega\})}{2}$$

$$\boxed{P_{abs.by.load} = 2.1195\{MW\}}$$

b) $$Q_{del.by.load} = \frac{-\left|\bar{I}_{bm}\right|^2 X_{LOAD}}{2} = \frac{-(530\{A\})^2(-50.719\{\Omega\})}{2}$$

$$\boxed{Q_{del.by.load} = 7.1235\{MVAR\}}$$

c) $$S_{abs.by.load} = P_{abs.by.load} + j\,Q_{abs.by.load}$$

$$\boxed{S_{abs.by.load} = (2.1195 - 7.1235j)\{MVA\}}$$

d) $$pf = \cos(-73.43°) = \boxed{0.28519 \text{ leading}}$$

It is leading because $\angle Z_{LOAD}$ is negative.

e) $$Z_{LINE} = (18 + 15.12j)\{\Omega\}$$

$$Y_{LINE} = \frac{1}{Z_{LINE}} = 42.539\{mS\}\angle -40.03°$$

$$= (32.572 - 27.361j)\{mS\}$$, so

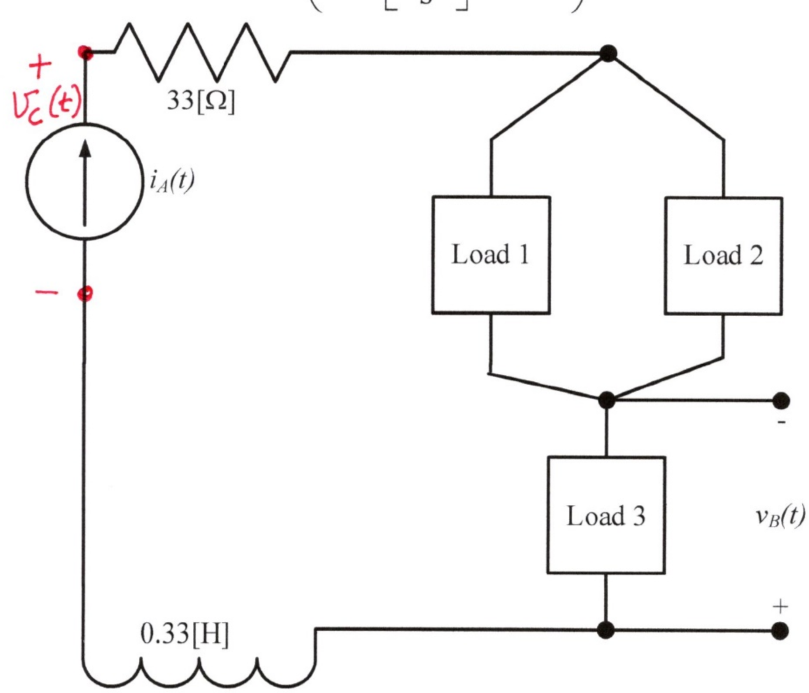
$$\boxed{B_{LINE} = -27.361\{mS\}}$$

5. (40 points) The circuit shown operates in steady-state. It is given that
Load 1 absorbs $(530\angle -23°)[\mathrm{VA}]$, and
Load 2 absorbs $390[\mathrm{W}]$, and delivers $250[\mathrm{VAR}]$.

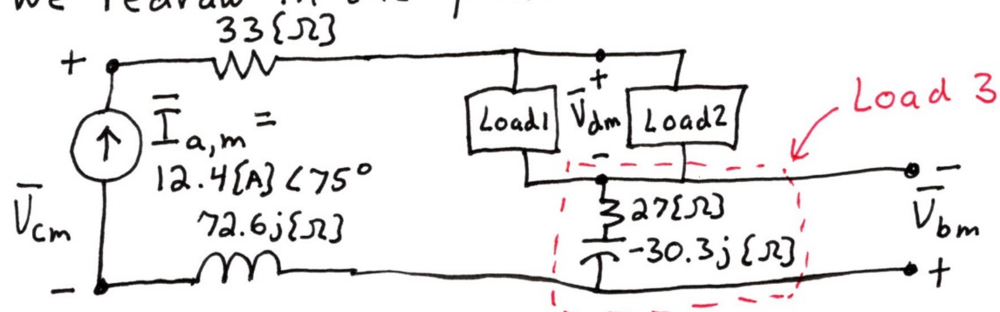
It is also given that Load 3 can be modeled by a resistor with a value of 27[Ω] in series with a capacitor with a value of 150[μF].

a) Find $v_B(t)$.
b) Find the voltage across the current source.
c) Find the power factor angle for the combination of the three loads, taken as a single load.

$$i_A(t) = 12.4[\mathrm{A}]\cos\left(220\left[\frac{\mathrm{rad}}{\mathrm{s}}\right]t + 75°\right)$$



We redraw in the phasor domain.



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a) $\bar{V}_{bm} = -\left(12.4\{A\}\angle 75^\circ\right)\left(27-30.3j\right)\{\Omega\}$

$= \left(503.25\angle -153.3^\circ\right)\{V\}$

so,

$$\boxed{v_B(t) = 503.25\{V\}\cos\left(220\left\{\tfrac{rad}{s}\right\}t - 153.3^\circ\right)}$$

b) We need the voltage across Load 1 and Load 2.

$S_{abs.by.Load\ 1} = \left(530\angle -23^\circ\right)\{VA\}$

and

$S_{abs.by\ Load\ 2} = \left(390-250j\right)\{VA\}$.

So

$$S_{abs.by.Loads\ 1+2} = S_{abs.by.Load\ 1} + S_{abs.by.Load\ 2} = \left(989.74\angle -27.505^\circ\right)\{VA\}$$

$$S_{abs.by.Loads\ 1+2} = \frac{\bar{V}_{dm}\cdot \bar{I}_{am}^{*}}{2} = \frac{\bar{V}_{dm}\left(12.4[A]\angle -75^\circ\right)}{2}$$

So, $$\bar{V}_{dm} = \frac{2\left(989.74\angle -27.505^\circ\right)\{VA\}}{12.4\{A\}\angle -75^\circ} = 159.64\{V\}\angle 47.495^\circ$$

KVL)

$$\bar{V}_{cm} = \bar{I}_{am}\left(33+72.6j\right)\{\Omega\} + \bar{V}_{dm} - \bar{V}_{bm} = 993.69\{V\}\angle 101.98^\circ$$

$$\boxed{v_c(t) = 993.69\{V\}\cos\left(220\left\{\tfrac{rad}{s}\right\}t + 101.98^\circ\right)}$$

see next page

Note that we made an arbitrary choice for the polarity of $v_c(t)$. If we had chosen the other polarity, that solution would also have been valid,

$$\boxed{993.69[\mathrm{V}]\cos\left(220\left[\tfrac{\mathrm{rad}}{\mathrm{s}}\right]t - 78.023^\circ\right)}$$

c) $S_{abs.by.Load\,3} = \dfrac{12.4[\mathrm{A}^2]}{2}(27-30.3j)[\Omega]$

$= (3.12\angle -48.296^\circ)[\mathrm{kVA}]$

$S_{abs.by.Loads\,1\text{-}3} = S_{abs.by.Load3} + S_{abs.by.Load\,1+2}$

$= 4.0599\angle -43.336^\circ[\mathrm{kVA}]$

Thus, the power factor angle = $\boxed{-43.336^\circ}$